# SYED ALI HUSSAIN

DevOps Engineer | Multi-Tenant SaaS • AWS & On-Prem • Identity • Automation

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## PROFILE

DevOps Engineer with production ownership of a multi-tenant B2B SaaS platform running across AWS (EC2, RDS, S3, Route 53, CloudWatch, Secrets Manager) and on-premise Ubuntu/Hyper-V infrastructure. Responsible for CI/CD pipeline design, disaster recovery architecture, secrets management, identity infrastructure (Active Directory), network security, and incident response across multiple backend services and client environments. Formal training in cybersecurity, cloud architecture, and DevOps through an EduQual UK-accredited RQF Level 6 diploma (Distinction, 88%). Comfortable across the stack — from writing Bash and Python automation, to designing multi-region DR strategy, to configuring routing on physical firewalls.

## CORE COMPETENCIES

**Cloud & IaC:** AWS (EC2, RDS, S3, VPC, IAM, CloudWatch, Lambda, SNS, Secrets Manager, EKS, Route 53), Azure (basic), Terraform, AWS Backup, cross-region disaster recovery

**CI/CD & Delivery:** GitHub Actions (self-hosted + GitHub-hosted runners), private container registries, immutable artifact promotion, atomic zero-downtime deploys, rollback strategies, composite actions, reusable workflows, Jenkins, Git

**Containers & Orchestration:** Docker, Docker Compose, Podman, Kubernetes (AWS EKS), RBAC, OPA policy enforcement, container hardening (non-root UIDs, image optimization, multi-stage builds)

**Databases & Caching:** PostgreSQL (multi-schema tenant isolation, backups, restore drills, connection pooling, replicas), Redis, AWS RDS, cross-region replication

**Identity & Access:** Active Directory (Windows Server, FSMO roles, Group Policy, AD-integrated DNS, dcdiag/repadmin), RADIUS/NPS, IAM policy design, least-privilege access, Laravel Passport, JWT

**Networking & Security:** MikroTik RouterOS (firewall rules, NAT, VLAN planning), dual-WAN failover with Netwatch, OpenVPN, WireGuard, Cloudflare (CDN, DNS, firewall rules), SSL/TLS (Certbot), Nginx rate limiting, incident response, secrets rotation, intrusion detection (Snort, Suricata)

**DNS & Email:** DNS record management (A, CNAME, MX, TXT, NS, SRV), Cloudflare & Route 53, email deliverability (SPF, DKIM, DMARC), corporate mail administration

**Virtualization & OS:** Hyper-V, WSL2, Windows Server, Linux (Ubuntu, Xubuntu, AlmaLinux), systemd, supervisorctl, cron, apt/dpkg, LVM, SELinux, UFW/iptables

**Automation & Scripting:** Bash, Python (argparse CLI utilities, OOP, requests, PEP 8), Ansible (playbooks, roles, Jinja2), n8n workflow automation, PowerShell

**Monitoring & Observability:** CloudWatch, Grafana, Prometheus, Loki, custom Bash-based monitors with webhook notifications, Nginx logs, SNS alerts, log rotation, centralized logging (rsyslog)

## PROFESSIONAL EXPERIENCE

**DevOps Engineer** — AIIT Solutions, Islamabad (Hybrid) *Jan 2026 – Present*

- Own the full infrastructure lifecycle for a multi-tenant B2B SaaS platform built as a Laravel distributed monolith (8 backend services + Vue 3 frontend, PostgreSQL multi-schema tenant isolation, Redis, S3) across DEV, DEMO (on-prem Hyper-V), UAT, and Production (AWS) environments.
- Cut the monthly AWS bill from ~$425 to ~$200 (~53%), including ~$79/month from EC2 and RDS right-sizing; cut per-service deploy time from 12 to 4 minutes with dependency and Docker layer caching; and cut recovery time from 1 h 25 min to 49 min with real-time replication, warm read replicas and standbys on AWS.
- Design and maintain CI/CD pipelines on GitHub Actions with self-hosted runners: build once via private container registry, promote the same immutable artifact through UAT to Production, atomic zero-downtime releases via timestamped release directories and symlink switching, with a tested rollback path.
- Led a cross-service CI/CD standardization initiative: consolidated duplicated deploy scripts into a reusable workflow, extracted notification logic into a composite action, added pre-deploy test gates and rollback capability, normalized branch triggers, and standardized container UIDs across the fleet.
- Architected the disaster recovery strategy: built a DR VPC in a secondary AWS region with private/public subnets and security groups, implemented S3 cross-region replication for production media (multi-GB, tens of thousands of objects), enabled Secrets Manager multi-region replication, and designed a three-tier recovery model (Primary → Colo cold site → AWS DR) with edge-driven failover.

- Managed a production Laravel authentication migration (Sanctum → Passport): designed environment-scoped key generation with AWS Secrets Manager storage, a fetch script pulled during CI/CD deploy on EC2, and coordinated the multi-service model where auth and superadmin both hold private keys for cross-service impersonation.
- Led incident response for a production queue outage: root-caused a supervisor worker misconfigured to consume only the default queue while critical jobs dispatched onto a named queue; fixed the configuration, added a missing scheduler supervisor program, and built a live monitoring stack (Bash + supervisorctl + PostgreSQL query monitors) posting alerts to the team collaboration channel every minute via crontab.
- Led incident response for a public-exposure security event on an internal platform: contained it at the edge firewall (office static IP plus VPN only), moved credentials to AWS Secrets Manager, added per-endpoint Nginx rate limiting, and verified response redaction before a controlled re-exposure.
- Migrated the office network from legacy to enterprise MikroTik infrastructure: firewall segmentation between on-prem, cloud, and external client traffic; reverse-proxy routing with double-NAT for external forwarding while preserving internal isolation; OpenVPN for remote developers; dual-WAN failover (primary + cold-standby ISPs) with Netwatch monitoring; remediated an 18-finding security audit.
- Administer a Windows Server Active Directory domain (single-DC forest, all 5 FSMO roles, AD-integrated DNS, Group Policy) supporting company-wide identity and access management. Evaluated MikroTik OpenVPN authentication via RADIUS/NPS against AD to replace per-user shared VPN secrets with domain credentials.
- Provisioned isolated network access for an external enterprise client's remote users: dedicated MikroTik port with its own subnet for the client's AD server, a dedicated WireGuard tunnel for their remote users, firewalled to authentication ports only — preserving on-prem network isolation while enabling secure external client access.
- Provisioned Dev/Staging/Production environments for an external enterprise client's Node.js/Next.js application on dedicated on-prem hardware: branch-per-environment strategy, isolated PostgreSQL databases per environment, systemd process management, and Nginx reverse proxy with Certbot SSL across multiple subdomains — delivered live and verified.
- Built cross-platform CI/CD for an internal .NET desktop application (Windows, Linux, macOS): hybrid GitHub Actions workflow using GitHub-hosted Windows runners and self-hosted Linux runners, ad-hoc code signing, and artifact distribution to a private storage bucket for internal QA.

**DevOps Intern** — Cloudlem (Pvt.) Ltd (Remote) *Nov 2025 – Dec 2025*

- Contributed to CI/CD pipeline configuration, system monitoring, and version-control workflows in a team-based environment, improving overall workflow efficiency and deployment reliability.
- Resolved persistent Linux file-permission conflicts between web servers and CI runners via ownership cascades and umask configuration, and scripted automated conflict-resolution logic into deployment pipelines.
- Recognized in an official Letter of Recommendation from the Founder & CEO for strong technical ability, problem-solving, and contribution to deployment-process improvements.

**DevOps Intern** — Al Nafi International College (Remote, Job Training Program of RQF Level 6 Diploma) *Jul 2024 – Jul 2025*

- Completed an intensive, hands-on job training program on the Al Nafi Cloud Platform (real-world IT infrastructure simulator), scoring 75–90% on complex lab assessments.
- Capstone training projects: Jenkins-based Docker deployment pipeline with S3 log archival; container-orchestrated web application on AWS EKS.

## CERTIFICATIONS & TRAINING

- **AWS Certified Solutions Architect** — Associate (in progress)
- **Certified Kubernetes Security Specialist (CKS)** — (in progress)

## EDUCATION

**Diploma in Artificial Intelligence Operations** — EduQual UK-accredited RQF Level 6 *Jul 2024 – Oct 2025*

*Al Nafi International College, Karachi • Overall Grade A • Distinction (88%)*

RQF Level 6 sits at the same level of the UK Regulated Qualifications Framework as an Honours Bachelor's degree (per QAA "Qualifications can cross boundaries" framework mapping). Curriculum: cybersecurity, cloud (AWS/Azure), DevOps toolchains (Terraform, Kubernetes, Prometheus/Grafana, Jenkins), Python programming, and AI/ML foundations. Assessed via automated lab reports on the Al Nafi Cloud Platform; scored 75–90% on complex labs.

**Intermediate (HSSC), Science General Group** — Adamjee Government Science College *Jul 2023 – Jul 2025*

*Board of Intermediate Education Karachi • Grade A (76.18%)*